DISCOVERING SEMI-PRIVATE SOCIAL NETWORKS: Structural Bias from Private–Private Edge Censoring

Dmitry ZINOVIEV* and Michael LI+

*Suffolk University, Boston, MA

+Wayland High School (student), Wayland, MA

PROBLEM

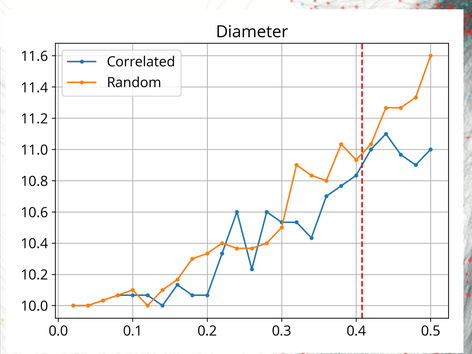
- An empirical network of East Slavic languages (RU, UA, BY) speakers in New England on Facebook
  - ~12,000 nodes
  - ~103,000 edges
- ~40% of users censor friend lists for privacy reasons
- Private users may remain visible through public neighbors, while private–private edges become hidden
- Research question: **How does hiding private–private edges distort network structure?**

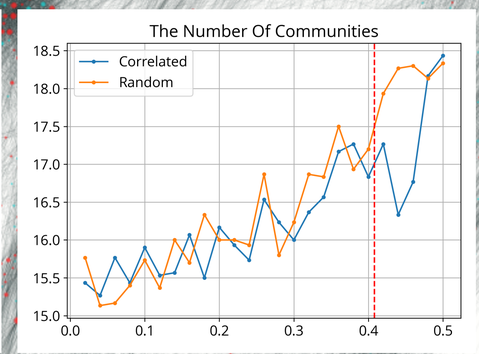
METHOD

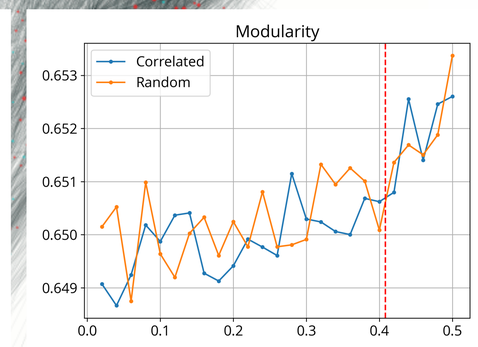
- Simulate the empirical network in two edge censoring modes
- Correlated mode: randomly declare some (N%) of the nodes private and eliminate the corresponding private–private (hidden) edges; repeat x30
- Random mode: randomly eliminate the same amount of edges regardless of the incident nodes' status (the **baseline**); repeat x30
- Calculate some network attributes: *clustering coefficient*, *giant connected component size* and *diameter*, *density*, the *number of Louvain communities*, and *Newman modularity*—and estimate the bias.

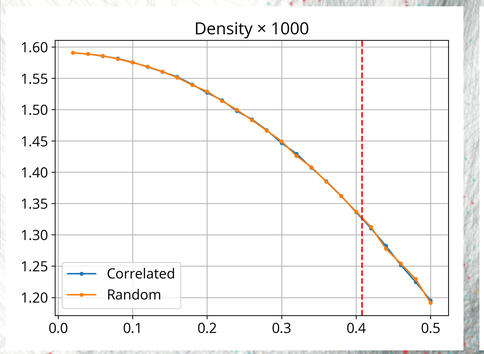
RESULTS (RANDOM PRIVATE NODE PLACEMENT)

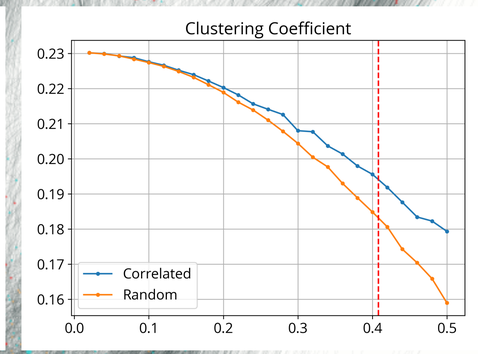
- The **observed** empirical network is an **estimate** of the real network
- The graphs on the right show the missing rate N as the X axis
- At the estimated privacy level of ~40% (the red line in the graphs), some global measures are relatively robust
- Overestimated in the observed network:
  - *diameter* (8%)
  - the *number of Louvain communities* (8%)
  - *Newman modularity* (0.2%)
- Underestimated in the observed network:
  - *density* (-20%)
  - *clustering coefficient* (-18%)
  - *GCC size* (-2%)
- The original network is likely denser, more locally clustered, slightly more globally connected, and more compact, with fewer communities and slightly weaker modular separation than the observed empirical network.

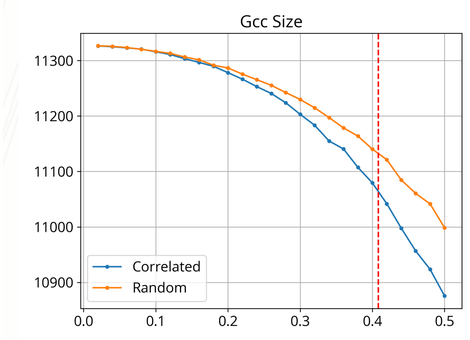






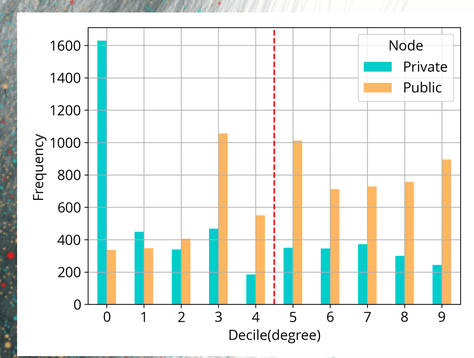






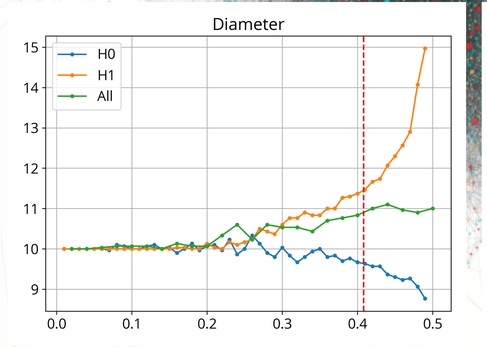
EFFECT OF PRIVATE NODE PLACEMENT

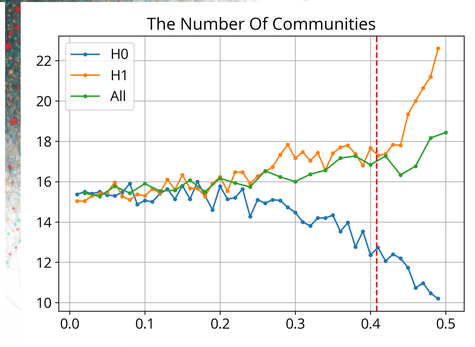
- Does the bias depend on the private node placement?
- The private nodes in the observed empirical network have smaller degrees; most of them are in the first decile
- Repeat the correlated experiment for the lower (H0) and upper (H1) halves of the degree distribution

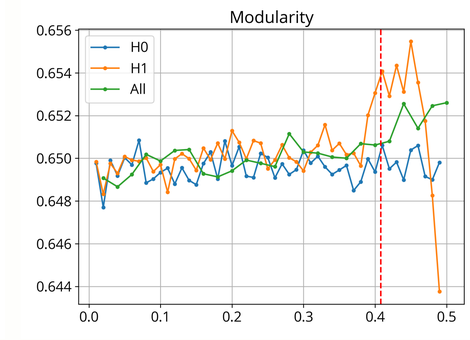


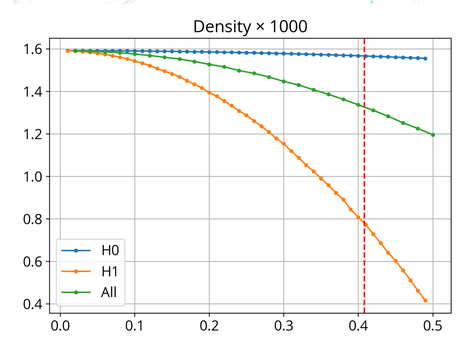
RESULTS (DEGREE-CONDITIONED)

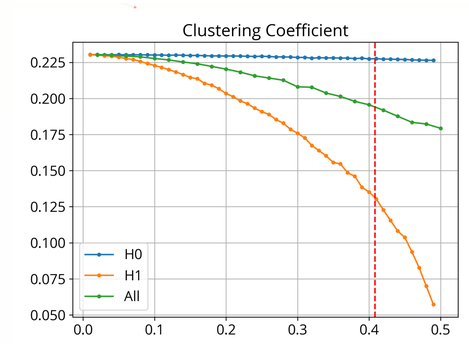
- The **observed** empirical network is an **estimate** of the real network
- The difference between correlated and random edge censoring has improved!
- Overestimated in the observed network:
  - *Newman modularity* (0.3%)
- Underestimated in the observed network:
  - the *number of Louvain communities* (-25%)
  - *diameter* (-3%)
  - *density* (-2%)
  - *clustering coefficient* (-1%)
  - *GCC size* (-1%)
- Under the lower-half degree selection model, the original network is slightly denser, slightly more connected, slightly less compact, and marginally more clustered than the observed network, but its main difference is **mesoscopic: it contains substantially more Louvain communities** with only slightly weaker modular separation.

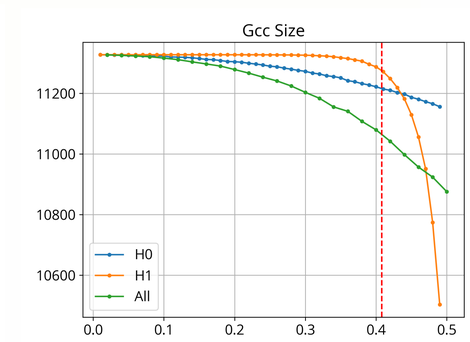












CONCLUSION

Discoverability in partially private social networks depends on privacy semantics, not merely missingness rates.

Modeling privacy semantics and private-node placement improves bias estimation for partially observed networks.

Reconstruction quality can be improved by accounting for the private node placement.

REFERENCES

[1] **Kossinets, G.** (2006). Effects of missing data in social networks. *Social Networks*, 28(3), 247–268

[2] **Smith, J. A., Moody, J., & Morgan, J. H.** (2017). Network sampling coverage II: The effect of non-random missing data on network measurement. *Social Networks*, 48, 78–99

[3] **Zheleva, E., & Getoor, L.** (2009). To join or not to join: The illusion of privacy in social networks with mixed public and private user profiles. In *Proceedings of the 18th International Conference on World Wide Web (WWW '09)* (pp. 531–540). Association for Computing Machinery

[4] **Borgatti, S. P., Carley, K. M., & Krackhardt, D.** (2006). On the robustness of centrality measures under conditions of imperfect data. *Social Networks*, 28(2), 124–136

[5] **Krause, R. W., Huisman, M., Steglich, C., & Snijders, T. A. B.** (2020). Missing data in cross-sectional networks. An extensive comparison of missing data treatment methods. *Social Networks*, 62, 99–112

[6] **Smith, J. A., Morgan, J. H., & Moody, J.** (2022). Network sampling coverage III: Imputation of missing network data under different network and missing data conditions. *Social Networks*, 68, 148–178